

Robin J

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EXPERIENCE

COMMERCEIQ | SOFTWARE DEVELOPMENT ENGINEER

Bangalore, India | Dec 2022 – Present

Key Achievements

- Unified the authentication and authorization systems across two sibling products by implementing a **unified password and SSO (Single Sign-On) login flow**. This solution, which had dependencies across more than 20 repositories, was designed as a scalable, plug-and-play system for future acquisitions.
- Led **BigQuery cost optimization**, reducing costs by over \$30,000/month by centralizing data from multiple location-based storage buckets into a single location. Improved query performance by auditing and seamlessly integrating with 10 different services, enhancing performance along the way.

Data Engineering & Optimization

- Improved backend API latency from over 1 minute to less than 5 seconds by **migrating the backend stack**—consumed by over 30 analytical dashboards with complex visualizations and data queries—from **Elasticsearch to BigQuery**, significantly enhancing speed, reliability, and scalability.
- Conducted an organization-level cost and **performance audit, optimizing several large BigQuery queries**. This resulted in an average speed improvement of 2x, with some queries optimized by more than 4x, while significantly reducing resource consumption.
- Worked extensively with **BigQuery and DBT** to transform several terabytes of raw data into rich, usable formats, improving performance for use across multiple teams.

Frontend & Backend Development

- **Migrated a legacy class-based React codebase** to React 18, adhering to the latest industry standards and improving code maintainability and performance.
- **Built multiple analytical dashboards** for complex data visualizations using React, writing both frontend and backend code in **Node.js** and BigQuery. Leveraged libraries such as Recharts, React-Data-Grid, and D3 for enhanced functionality.
- **Developed self-service tools** for configuration management with features like **asynchronous bulk updates, role-based access control, and error handling**, improving management efficiency and reducing misconfiguration.
- Created BI (Business Intelligence) dashboards for clients and internal users using **Looker**, enabling **improved metrics reporting**, transparency, and data-driven decision-making.

Cloud & Infrastructure

- Migrated several services, including the primary frontend and backend **CI pipelines, from Jenkins to GitHub Actions (GHA)**. This reduced deployment time from 40 minutes to less than 5 minutes by implementing best practices such as caching, resource sharing across steps, and parallelizing jobs.
- Designed and built infrastructure for asynchronous report exports using **Cloud Tasks, Cloud Run, and Terraform**, ensuring scalability and reliability. This system included **automated email notifications** upon job completion.
- Planned and wrote the **infrastructure for monitoring data reliability, automatically creating Jira tickets** for anomalies exceeding defined thresholds. This reduced data discrepancies and facilitated earlier issue identification.

Collaboration & Knowledge Sharing

- Authored detailed documentation on products, features, and key projects, significantly improving onboarding time for new hires and supporting teams with runbooks.
- Assisted teammates with complex SQL queries and debugging, fostering a collaborative work environment.
- Mentored new team members, onboarding them to BigQuery, GCP, and general development best practices.

- Migrated a part of the database from **ElasticSearch** to **Clickhouse**.
- Wrote a **query builder** to dynamically build SQL queries from conditions that can be dynamically set from the UI.
- Came up with the optimal data structures to store OLAP data in Clickhouse which we ended up using.
- Tackled the various **OLAP** update/delete problems efficiently.
- These reduced the query time from not-usable to under 4 seconds.
- Worked with Kafka Stream data. Wrote a service that listens to **Kafka** stream and writes to the database after needed data transformations.

FREELANCE PROJECTS | SOFTWARE DEVELOPER

Tamil Nadu, India | Aug 2020 – Jan 2022

- **ARIMA LABS (Aug 2021 – Dec 2021):** Developed a traffic visualization dashboard using React and TypeScript, with a **Django backend and MySQL database**. Redesigned database schema, implemented pagination, and optimized queries, improving speed by 10x. Built Node.js infrastructure for image storage and **RTSP-to-HTTP video stream** conversion, reducing RAM usage by 50
- **HOODS (Oct 2021 – Jan 2022):** Created a social media app in **Flutter with Firebase** for local communities. Implemented Firebase Cloud Messaging, Cloud Functions, and Realtime DB integration with Firestore to track user activity. Used ObjectBox caching to reduce bandwidth by 30% and employed GetX for state management.
- **AMMAMMA'S (Aug 2020 – Nov 2020):** Designed and launched an E-Commerce app using Flutter and Firebase, featuring user reviews, online payments (via **Stripe**), and an admin panel for order management. Built a separate delivery tracking app and released the main app on the Play Store, achieving 1,000+ downloads and a 4.6-star rating.

OTHER PROJECTS

ALGORITHMS VISUALIZER

REACT, DSA

A visualizer built with ReactJs to visualize various **Path finding** and **Sorting** algorithms. Like Dijkstra's algorithm, A* algorithm, Merge Sort, Quick Sort, Heap Sort, etc.

SPOTIFY LYRICS FINDER

FLUTTER, FIREBASE, SPOTIFY API, GENIUS API, AUTHENTICATION, CACHING

A mobile application built using Flutter to get the lyrics of the song you are currently listening to on Spotify. I've used Spotify OAuth to connect to the Spotify account. The lyrics are fetched from Genius API.

FLAPPY BIRD

P5JS, CSS, HTML

A simple Flappy bird game built using HTML and p5js.

EDUCATION

BE. Computer Science Engineering

9.36 CGPA | Erode, TN, India

KONGU ENGINEERING COLLEGE

SKILLS

Languages: Javascript, Typescript, Python, Golang, Java, SQL, Ruby

Frameworks: Flutter, React, Nodejs, NextJs, ExpressJs, Django, Ruby on Rails, ElectronJs

Databases: Postgresql, BigQuery, Firestore, MongoDB, Clickhouse

Tools & Platforms: Looker, DBT, Spark, Terraform, GitHub Actions, Jenkins

Cloud: GCP (Cloud Run, CloudTasks, Kubernetes), AWS (EC2, S3)

Other: Firebase, Git, Kafka, CI/CD, Docker, SQL optimization